Laboratory 1:

Signals and Systems
CT/DT Signal Pipeline Bring-Up for Embedded Prototypes

Arduino Signal Generator & Cross-Domain Measurement Pipeline

Lab Duration: 2 hours
Pre-Lab Time Expected: 30 minutes (filter design and jitter analysis)
Key Concept: Translating continuous-time signal theory into robust discrete-time embedded implementations and validating signal fidelity across oscilloscope and software acquisition domains.

Contents

Grading and Authenticity Requirements

Deliverable (1/16): This is a graded laboratory assignment. Include your **name and student ID** in all required screenshots and photographs (oscilloscope, MATLAB plots, breadboard photos).

Deliverable (2/16): All waveforms and jitter measurements must be acquired from your own hardware build and software implementation. Pre-computed synthetic data or borrowed results are not accepted.

Checkpoint (1/8): Before submission, verify that each MATLAB figure includes axis labels with units, a grid, a title with your name/ID, and a legend (if applicable).

Title & Real-World Context

You are a Junior Embedded Systems Engineer at a Tier-1 IoT/Sensing Firm

Your technical lead has tasked you with a **critical bring-up and validation** for a microcontroller signal-generation and acquisition pipeline that powers three concurrent commercial programs:

1. **TinyML Edge Inference Platform.** A medical-device partner is deploying real-time arrhythmia detection on a sub-1-mA wearable processor. Your platform must generate low-power, **phase-coherent** test signals to stress the analog front-end and validate that the feature-extraction pipeline (QRS detection, HRV filtering) meets sub-100ms latency and 98% sensitivity thresholds. The challenge: strict SRAM constraints (32 KB on your target processor) and competing interrupt budgets demand that you benchmark whether algorithmic waveform generation (CPU-intensive `sin()`) or pre-computed lookup tables (memory-intensive PROGMEM) yields better deterministic timing. Poor choices here invalidate the entire model-validation chain.

2. **Aerospace Stratospheric Intelligence Platform.** A high-altitude imaging firm is deploying sensor pods (1–3 hour flight envelope) to gather multi-spectral atmospheric data. Your signal pipeline must emulate the analog sensors' behavior and validate that the spacecraft's onboard data logger captures, timestamps, and downlinks raw waveforms with <1% time-drift over mission duration. Any jitter, aliasing, or quantization loss directly cascades into downstream ground-station imagery processing and can render science data unusable. You will measure this via oscilloscope cross-correlation and MATLAB statistical analysis.

3. **Industrial IoT Vibration Monitoring.** A predictive-maintenance startup uses mill-floor accelerometers to detect bearing fatigue before catastrophic failure (saving $500k+ downtime per incident). Their edge processor must reliably generate diagnostic test signals and acquire real vibration spectra with minimal phase distortion and anti-aliasing. Aliasing artifacts or sampling irregularities that corrupt higher harmonics (2–10 kHz band) would mask early fault signatures and defeat the entire condition-monitoring premise.

Your Mission. In this lab, you will **architect and validate** a complete signal pipeline from discrete-time mathematics to analog reconstruction:

- Design a passive anti-imaging filter that removes PWM carrier harmonics while preserving signal fidelity.
- Synthesize waveforms using either algorithmic or lookup-table approaches and quantify memory-versus-timing trade-offs.
- Acquire signals via oscilloscope (analog ground truth) and MATLAB serial capture (software validation layer).
- Compare and rank error sources (quantization, jitter, component tolerance, filter rolloff) by magnitude and impact on downstream models.

Industry teams obsess over this workflow because **a single ill-chosen parameter** (wrong RC cut-off, excessive jitter variance, serial baud-rate aliasing) can silently corrupt months of field deployment or flight test data. Your systematic approach to measurement, error analysis, and hardware-software cross-validation demonstrates the rigor required for mission-critical systems.

Constraints (Fixed). 2-hour lab execution window, existing hardware platform (Arduino Uno,

oscilloscope, MATLAB). All skeleton code templates provided to keep focus on core signal integrity concepts rather than boilerplate USB and timer configuration. All technical checkpoints and deliverables remain exactly as specified herein.

1 Learning Objectives

Upon completion of this laboratory, you will demonstrate:

1. **Precision Waveform Synthesis Under Resource Constraints.** Make informed trade-off decisions between algorithmic signal generation (CPU-intensive, memory-light) and lookup-table methods (memory-intensive, timing-optimal), and quantify impact on total system jitter and inference/sensing accuracy.
2. **Anti-Aliasing Filter Design and Empirical Validation.** Select RC component values to achieve target cutoff frequency, build a passive reconstruction stage, measure frequency response via oscilloscope, and verify that PWM harmonics are adequately suppressed to prevent aliasing of baseband content.
3. **Cross-Domain Measurement Integrity.** Design and execute a dual-validation measurement strategy—simultaneous oscilloscope analog capture and digital MATLAB serial acquisition—to expose systematic errors and validate that your platform meets signal-fidelity requirements for safety-critical downstream processing.
4. **Jitter Characterization and Deterministic-Timing Optimization.** Measure timing variance in both software-loop and hardware-interrupt architectures, quantify the performance gap using statistical analysis, and justify your choice of implementation strategy for your application domain (edge ML, aerospace, or vibration monitoring).

2 Core Concepts: Discrete-Time Waveform Synthesis and Analog Reconstruction

Digital Signal Processing dictates the transformation of continuous-time events $x(t)$ into discrete-time sample arrays $x[n]$. A microcontroller operates inherently in discrete time governed by a local clock. Therefore, waveform synthesis relies upon discretizing mathematical relationships bounded by fundamental constraints such as the Nyquist-Shannon Sampling Theorem.

2.1 PWM-to-Analog Filtering Architecture

Pulse Width Modulation (PWM) encodes arbitrary analog voltages by dynamically modifying the duty cycle D of a digital pulse train oscillating at a fixed carrier frequency f_{pwm}. The logic-high width T_H characterizes the instantaneous equivalent continuous voltage:

$$v_{out}(t) = \left(\frac{T_H}{T_{pwm}}\right) V_{ref} = D \cdot V_{ref} \tag{1}$$

Because the signal contains large spectral components localized around f_{pwm} and its harmonics, a passive anti-imaging low-pass filter is required to extract the pure baseband continuous signal. The fundamental cutoff frequency f_c of a first-order RC filter is defined as:

$$f_c = \frac{1}{2\pi RC} \tag{2}$$

The parameter space limits f_c to be sufficiently larger than the baseband signal frequency f_0 yet far attenuating compared to the carrier frequency f_{pwm} ($f_0 < f_c \ll f_{pwm}$).

2.2 Digital Discretization and The Sampling Frequency Constraint

To prevent spectral folding (aliasing), the discrete sampling interval T_s must satisfy the continuous counterpart boundary conditions:

$$x[n] = x(nT_s) \Rightarrow f_s = \frac{1}{T_s} \geq 2f_{max} \tag{3}$$

For internal memory look-up-tables within the microcontroller, T_s is controlled via processor loops or dedicated hardware timer interrupts. Interrupt-driven updates limit systemic loop-delay variance (jitter) compared to sequential polling methodologies.

3 Pre-Lab Assignment

Provide rigorous mathematical derivation for the following problems.

1. **Pre-lab Deliverable (1/3):** Filter Design Parameters:Select commercially available R and C for $f_c \approx 50$ Hz using $f_c = \frac{1}{2\pi RC}$.
2. **Pre-lab Deliverable (2/3):** Impulse Response:Derive $h(t)$ and calculate time constant $\tau = RC$ for 63% settling.
3. **Pre-lab Deliverable (3/3):** Quantization Analysis:Calculate quantization voltage step ΔV and SQNR (dB) for 8-bit PWM.

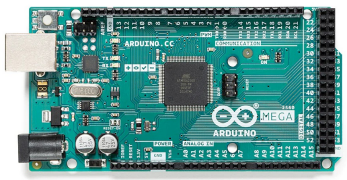
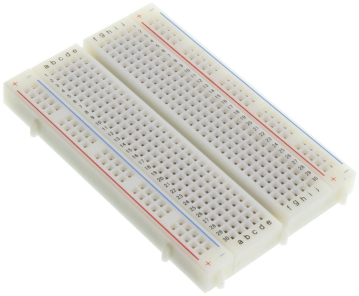
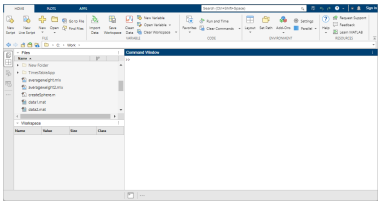
4 Equipment & Setup

Table 1 bounds the specific components used for this pipeline framework.

Table 1. Hardware and DAQ Software Requirements

Classification	Required Tooling/Resources
Compute System	Arduino Microcontroller (Uno or equivalent), Standard USB A-to-B Serial bus.
Analog Synthesis	Prototyping Breadboard, Basic Resistor/Capacitor kits, Jumper routing wires.
Bench Instrumentation	High-resolution Digital Oscilloscope or ADALM2000 hardware suite.
IDE & Processing	Arduino Studio IDE (C++ deployment), MATLAB (Serial Acquisition Tooling).

Table 2. Visual Reference of Key Equipment

Arduino Uno	**USB cable**	**Digital Oscilloscope**
		
Breadboard	**Resistor/Capacitor Kits**	**MATLAB Interface**
		

5 Laboratory Procedure

5.1 Phase 1: Hardware Assembly and Basic Code Verification

5.1.1 Step 1.1: Design and Assemble RC Filter

Checkpoint (2/8): Implement the Anti-Imaging Filter:Using your Pre-Lab calculations, construct the first-order RC low-pass filter on the breadboard using the specified resistor and capacitor values.

Procedure: Build RC filter using Pre-Lab values. Measure actual components with multimeter. Include photo in lab report.

Deliverable (3/16): Submit breadboard photo with labeled resistor, capacitor, Arduino pin connections, and measured component values.

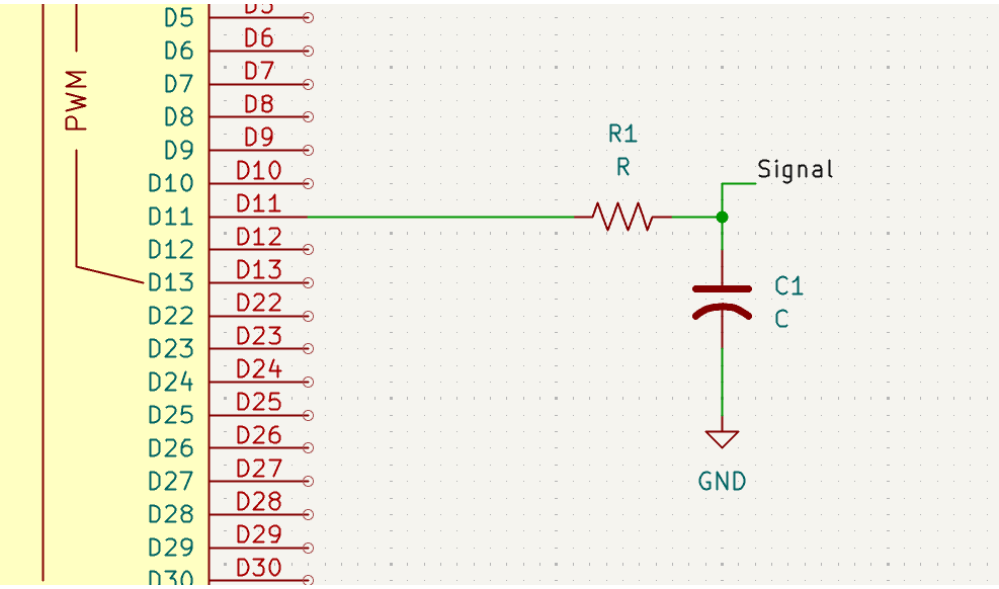


Figure 1. **REFERENCE SCHEMATIC:** Include breadboard photo and measured component values in lab report.

Deliverable (4/16): Your Circuit Documentation: Include breadboard photo with labeled resistor, capacitor, and Arduino connections in the lab report.

5.1.2 Step 1.2: Baseband Verification Test

Checkpoint (3/8): Baseband Verification:Verify that the filter produces expected DC voltage levels corresponding to PWM duty cycles.

Procedure: Test PWM at 0%, 50%, and 100% duty cycles using DMM. Verify filter output is linear (0V $\approx$ 2.5V $\approx$ 5V). Document results.

Deliverable (5/16): Submit DMM verification evidence for 0%, 50%, and 100% duty-cycle measurements.

Example Arduino Code:

```
pinMode(3, OUTPUT);
analogWrite(3, 0);    // 0% duty, measure output
delay(500);
analogWrite(3, 127);  // 50% duty, measure output
delay(500);
analogWrite(3, 255);  // 100% duty, measure output
```

Deliverable (6/16): Your DMM Photo: Photograph your digital multimeter display showing the voltage reading at one of the duty cycle settings (0%, 50%, or 100%). Show your Arduino code on screen if possible, or write it next to the photo.

Discussion (1/7): Compare expected vs. measured DMM voltages at 0%, 50%, and 100% duty cycle. Explain any nonlinearity or offsets using component tolerance, PWM ripple, or measurement setup.

5.2 Phase 2: Signal Acquisition and MATLAB Integration

5.2.1 Step 2.1: Waveform Generation

Checkpoint (4/8): Waveform Generation Matrix:Extend your Arduino code to generate sine, square, and triangle waveforms at your assigned frequency.

Deliverable (7/16): Submit your working waveform-generation code (LUT/algorithmic/hybrid) with timing parameter documentation.

Your Assigned Parameters:

Implementation approaches: (Choose one or more)

- **Look-Up Table (LUT):** Pre-compute 256 values stored in PROGMEM
- **Algorithmic:** Real-time computation using `sin()`, `cos()`, or PWM math
- **Hybrid:** Combine both methods for performance

REFERENCE - Example Sine Wave Implementation:

```
// Method 1: Look-Up Table (LUT)
const byte sine_lut[256] PROGMEM = {128, 131, 134, ...}; // 256 values
void setup() { /* initialize */ }
void loop() {
  for(int i = 0; i < 256; i++) {
    byte val = pgm_read_byte(&sine_lut[i]);
    analogWrite(PWM_PIN, val);
    delayMicroseconds(period_us);
  }
}

// Method 2: Real-time Algorithm
void loop() {
  for(int i = 0; i < 256; i++) {
    float sine_val = 128 + 127*sin(2*PI*i/256);
    analogWrite(PWM_PIN, (byte)sine_val);
    delayMicroseconds(period_us);
  }
}
```

Key Parameters: $T_s = \frac{1}{256 \times (\text{frequency in Hz})}$
For $f_0 = 10$ Hz: $T_s \approx 390$ μs per sample

Figure 2. **REFERENCE IMPLEMENTATIONS:** Include your working waveform generation code in lab report with timing documentation.

5.2.2 Step 2.2: Oscilloscope Capture and Verification

Checkpoint (5/8): Oscilloscope Bench Capture:Connect the filter output to the oscilloscope and capture your waveform.

Procedure:

Capture 3-4 complete periods showing clear waveform with stable trigger. Record time/div and volts/div settings. Photograph screen showing frequency, amplitude, and any distortion.

Deliverable (8/16): Submit oscilloscope screenshot/photo with visible time/div, volts/div, measured frequency, and waveform stability.

REFERENCE - Oscilloscope Display Layout:

Scope Capture Here

Key Measurements to Record:

- Read **Frequency** using vertical reference lines to measure period T
- Measure **Peak-to-Peak voltage** by counting vertical divisions
- Verify trigger is stable (no drift or jitter in waveform display)

Figure 3. **REFERENCE MEASUREMENT GUIDE:** Include scope photo in lab report with frequency, amplitude, and stability assessed.

5.2.3 Step 2.3: MATLAB Serial Data Acquisition

Checkpoint (6/8): MATLAB Serial Cross-Domain Pipeline:Acquire 500 samples of your waveform through serial communication.

Procedure:

Configure Arduino for serial transmission (115200 baud). Write MATLAB script to acquire 500 contiguous samples. Calculate actual sampling period T_s and frequency $f_s = 1/T_s$. Document results.

Deliverable (9/16): Submit MATLAB serial-acquisition script and Arduino transmitter code evidence (screenshots or code snippets).

REFERENCE - MATLAB Serial Communication Template:

```
% Configure serial port
clear; clc;
port = 'COM3';          % Change to your port
baud = 115200;
timeout = 60;           % seconds

% Create serial object
s = serialport(port, baud);
configureTerminator(s, "CR/LF");

% Read data
N_samples = 500;
data = [];
tic;
while length(data) < N_samples
    if s.NumBytesAvailable > 0
        val = readline(s);
        data = [data; str2double(val)];
    end
    if toc > timeout
        disp('Timeout!');
        break;
    end
end

% Process data
Ts = 1/Fs;  % sampling period (calculate from timing)
t = (0:length(data)-1) * Ts;
V = (data / 1023) * 5;  % convert ADC to volts

% Plot
figure; plot(t, V, 'b', 'LineWidth', 1.5);
xlabel('Time (s)'); ylabel('Voltage (V)');
title('Serial Acquisition at YOUR_FREQ Hz');
grid on;
```

Your Arduino Code (Serial Transmission):

```
void setup() { Serial.begin(115200); }
void loop() {
  int adc_val = analogRead(A0);
  Serial.println(adc_val);
  delayMicroseconds(period_us);
}
```


Figure 4. **REFERENCE CODE TEMPLATES:** Include screenshots of your working Serial.read() and analogRead() code in lab report.

5.2.4 Step 2.4: Data Visualization

Deliverable (10/16): Plot the buffered vectors:Create publication-quality plots of your acquired data.

Procedure: Create time vector: $t = [0 : T_s : (N-1) \cdot T_s]$ where $N = 500$. Scale ADC to voltage: $V = \frac{\text{ADC}}{1023} \times 5\text{V}$. Plot with labels, grid, and title. Include screenshot in lab report.

REFERENCE - Expected MATLAB Plot Structure:

MATLAB Plotting Code:

```
figure('Position', [100 100 800 400]);
plot(t, V, 'b-', 'LineWidth', 1.5);
xlabel('Time (s)', 'FontSize', 12);
ylabel('Voltage (V)', 'FontSize', 12);
title(['Serial Data: ' num2str(measured_freq) ' Hz Sine Wave'], ...
    'FontSize', 14);
grid on;
legend('Arduino ADC Reading', 'Location', 'northeast');
ylim([0 5]);
```

Figure 5. **REFERENCE PLOT STRUCTURE:** Include screenshot of your MATLAB figure (500 samples) with time, voltage, title, and grid in lab report.

Discussion (2/7): Compare your scope waveform and MATLAB waveform for frequency, amplitude, and distortion. Identify the dominant source of mismatch between analog capture and serially sampled data.

5.3 Phase 3 (The Challenge): Interrupt Optimization

5.3.1 Step 3.1: Software-Based (Baseline) Assessment

Before implementing interrupt-driven sampling, document the jitter characteristics of standard delay-based code:

Procedure:

Record loop timestamps using `micros()`. Calculate inter-sample intervals and compute mean, std dev, min, max. Document in lab report with histogram.

Deliverable (11/16): Submit baseline (delay-based) jitter histogram and summary statistics (mean, std dev, min, max).

REFERENCE - Jitter Measurement Code (Baseline):

```
// Arduino: Measure delay() loop jitter
void loop() {
  static unsigned long prev_time = 0;
  unsigned long curr_time = micros();

  if (prev_time != 0) {
    unsigned long interval = curr_time - prev_time;
    Serial.println(interval);  // Send to PC
  }

  prev_time = curr_time;
  analogWrite(PWM_PIN, next_value);
  delay(10);  // or whatever period needed
}
```

MATLAB: Histogram of Inter-Sample Intervals

```
% Read intervals from Arduino serial, then:
figure;
histogram(intervals_us, 50, 'FaceColor', [0.3 0.6 0.8]);
xlabel('Inter-Sample Interval (us)');
ylabel('Frequency (counts)');
title('Phase 2 Baseline: Delay-Based Jitter Analysis');
grid on;

% Statistics
mean_interval = mean(intervals_us);
std_interval = std(intervals_us);
fprintf('Mean: %.1f us, Std Dev: %.1f us\n', ...
    mean_interval, std_interval);
```

Expected Result: Histogram shows wide distribution (high jitter) around target interval, with standard deviation typically 5-50

Figure 6. **REFERENCE MEASUREMENT CODE:** Include histogram of inter-sample intervals (Phase 2 baseline) in lab report showing wide distribution.

5.3.2 Step 3.2: Hardware Interrupt Refactoring

Checkpoint (7/8): Hardware Interrupt Refactoring:Implement timer-driven sampling using hardware interrupts.

Procedure: Configure Timer1 for 2× Phase 2 sampling rate. Implement Interrupt Service Routine (ISR) for ADC reads. Remove all `delay()` calls from sampling. Verify waveform quality. Include code in lab report.

Deliverable (12/16): Submit interrupt-driven implementation code (timer setup + ISR) with final sampling-rate settings.

REFERENCE - Timer1 Interrupt Implementation (Arduino Uno):

```
// Global variables
volatile byte sample_index = 0;
const byte sine_lut[256] PROGMEM = { /* 256 values */ };

void setup() {
  Serial.begin(115200);
  pinMode(3, OUTPUT);

  // Configure Timer1 for interrupt-driven sampling
  noInterrupts(); // Disable interrupts

  TCCR1A = 0;
  TCCR1B = 0;
  TCNT1 = 0;

  // Set compare value (adjust for desired frequency)
  // For 10 Hz with 256 samples: OCR1A ~= F_CPU / (256*10*prescale)
  OCR1A = 6249;  // Example for 10 Hz, prescaler=8

  TCCR1B |= (1 << WGM12);    // CTC mode
  TCCR1B |= (1 << CS11);     // Prescaler=8
  TIMSK1 |= (1 << OCIE1A);   // Enable Compare A interrupt

  interrupts(); // Enable interrupts
}

// Interrupt Service Routine (ISR)
ISR(TIMER1_COMPA_vect) {
  byte val = pgm_read_byte(&sine_lut[sample_index]);
  analogWrite(3, val);
  sample_index = (sample_index + 1) % 256;
}

void loop() {
  // Main code - NO sampling here, ISR handles it!
  Serial.println("Interrupt-driven sampling active");
  delay(1000);
}
```

Key Points:

- ISR must be SHORT (microseconds execution)
- Use `volatile` for variables shared between ISR and main code
- Prescaler and OCR1A determine interrupt frequency
- NO `delay()` or `Serial` calls inside ISR

Figure 7. **REFERENCE TIMER1 CONFIGURATION:** Include your ISR implementation (2× frequency) in lab report with register settings explained

5.3.3 Step 3.3: Sampling Rate Augmentation

Checkpoint (8/8): Sampling Rate Augmentation:Double your sampling frequency using the interrupt mechanism.

Deliverable (13/16): Submit Phase 2 vs. Phase 3 jitter comparison (histograms) and computed improvement percentage.

Original Parameters (Phase 2):

Record Phase 2 baseline ($f_{s,\text{orig}}$, $T_{s,\text{orig}}$). Double frequency to $f_{s,\text{new}} = 2 \times f_{s,\text{orig}}$. Measure Phase 3 jitter (mean, std dev, min, max intervals). Calculate improvement factor. Document all results.

REFERENCE - Jitter Comparison Concept:

Histogram Comparison Here

MATLAB Code to Generate Comparison:

```
% Assuming intervals_phase2 and intervals_phase3 measured
figure;

% Compare histograms
subplot(1,2,1);
histogram(intervals_phase2, 50, 'FaceColor', [1 0.5 0]);
xlabel('Interval (us)'); title('Phase 2: Delay-Based Jitter');
xlim([min(intervals_phase2)*0.9, max(intervals_phase2)*1.1]);

subplot(1,2,2);
histogram(intervals_phase3, 50, 'FaceColor', [0 0.7 1]);
xlabel('Interval (us)'); title('Phase 3: Interrupt-Driven Jitter');
xlim([min(intervals_phase2)*0.9, max(intervals_phase2)*1.1]);

% Calculate improvement
jitter_phase2 = std(intervals_phase2);
jitter_phase3 = std(intervals_phase3);
improvement = (jitter_phase2 - jitter_phase3) / jitter_phase2 * 100;
fprintf('Jitter Improvement: %.1f%%\n', improvement);
```

Figure 8. **REFERENCE COMPARISON CODE:** Include side-by-side histograms (Phase 2 vs Phase 3 jitter) in lab report showing improvement percentage.

Discussion (3/7): Explain why timer-interrupt scheduling reduces jitter compared to delay-based loops. Discuss at least one tradeoff (complexity, ISR constraints, or resource usage).

6 Data Analysis

Complete verification of continuous representation using the captured discrete mapping pairs.

6.1 Analysis Step 1: Amplitude Verification

Discussion (4/7): Analog Authenticity Verification:Compare expected versus measured amplitudes across all three capture domains.

Theoretical vs. Experimental Comparison:

Discussion (5/7): Amplitude Verification:Compare theoretical vs. measured (scope and MATLAB): Calculate percent error $\epsilon = \frac{|A_{\text{theory}} - A_{\text{meas}}|}{A_{\text{theory}}} \times 100\%$ for both domains. Explain major discrepancies.

6.2 Analysis Step 2: Frequency Verification

Discussion (6/7): Frequency Verification:Compare programmed f_0 vs. measured (scope and MATLAB). Calculate frequency error: $\epsilon_f = \frac{|f_0 - f_{\text{meas}}|}{f_0} \times 100\%$. Expected $< 10\%$.

6.3 Analysis Step 3: Jitter Impact Assessment

Discussion (7/7): Jitter Impact:Compare Phase 2 (software delay-based) vs. Phase 3 (interrupt-driven) jitter measurements. Calculate reduction percentage: $R = \frac{\sigma_2 - \sigma_3}{\sigma_2} \times 100\%$. Target $30 - 80\%$ improvement.

6.4 Analysis Step 4: Digital-Analog Mapping Proof

Deliverable (14/16): Digital-Analog Mapping (LLM-Proof Integrity Requirement):Provide dual-platform evidence of authentic waveform acquisition.

Required Documentation:

Deliverable (15/16): Proof Documentation:Submit as separate JPG/PNG files in lab report:

- **Scope Photo:** Clear waveform, **handwritten name + ID visible**, time/div, volts/div settings shown
- **MATLAB Plot:** 500-sample time series, proper axes (time in seconds, voltage 0-5V), title with frequency, grid on
- Verify scope and MATLAB show same waveform characteristics (frequency, amplitude)

6.5 Analysis Step 5: Theoretical vs. Systemic Losses Summary Table

Deliverable (16/16): Submit summary table with theoretical vs. measured values and ranked error sources with justification.

Create summary table in lab report showing theoretical vs. measured values with percent error for amplitude, frequency, and period. Identify primary error sources: component tolerances, ADC quantization, PWM ripple, jitter, signal losses. Prioritize errors by magnitude.

7 Grading Rubric

Total Points: 100

Table 3. Detailed Grading Rubric with Specific Criteria

Category	Evaluation Criteria	Weight
Pre-Lab (15 pts)	Q1: RC filter values w/ f_c calculation ($\pm 10\%$)	5
	Q2: Impulse response $h(t)$ and $\tau = RC$	5
	Q3: Quantization ΔV and SQNR formula	5
Hardware (12 pts)	RC filter constructed & verified ($\leq 5\%$ error)	6
	DMM photos at 0%, 50%, 100% duty cycles	6
Scope Capture (10 pts)	Photo: 3-4 periods, frequency/amplitude $\leq$ 10% error	5
	Visible settings (time/div, volts/div), stable trigger	5
MATLAB Acquisition (10 pts)	500 samples acquired, time/voltage axes correct	5
	Amplitude within 10% of scope, grid & title included	5
Phase 3 Interrupt (12 pts)	ISR implemented, 2× frequency achieved	6
	Jitter reduction $\geq 30\%$, blocking calls removed	6
Proof Documentation (21 pts)	Scope photo: legible name/ID, waveform clear, settings visible	10
	MATLAB plot: time/voltage axes, grid, title, matches scope	11
Analysis & Errors (20 pts)	Error calculations & comparison table (amplitude, freq, period)	10
	Identification & ranking of 3+ error sources with justification	10
Totals		**100**

References and Inspiration

This laboratory manual is part of a comprehensive signal processing curriculum redesign that integrates pedagogical innovations from leading universities with real-world industry applications.

University Course Inspirations

Georgia Tech ECE 2026: Real-Time DSP with Embedded Hardware

Lab 01 draws direct inspiration from Georgia Tech's flagship signals and systems course, which emphasizes embedded signal generation on Arduino and microcontroller platforms. The use of PWM for tone generation, interrupt-driven signal synthesis, and real-time parameter adjustment reflects Georgia Tech's philosophy of making signal processing tangible through immediate hardware feedback. Students see their code produce audible and visible results within minutes, building intuition for frequency, amplitude, and phase relationships.

Rice University ELEC 241: Embedded Real-Time Systems

Rice's approach to embedded signal processing on resource-constrained platforms (initially TI LaunchPad, now Arduino-compatible) informs the emphasis on interrupt-driven architecture and timer-based precision. Lab 01's focus on jitter reduction and transition from polling to interrupt-driven code mirrors Rice's hands-on exploration of real-time constraints in embedded DSP.

MIT 6.003 / 6.004: Signals and Digital Systems

MIT's foundational treatment of signal properties (frequency, phase, amplitude) and digital systems provides the theoretical backbone. The pre-lab calculations and validation methodology (comparing oscilloscope observations to mathematical predictions) reflect MIT's commitment to rigorous mathematical thinking paired with physical verification.

Duke University ECE 280: Signals and Systems Lab Series

This lab is part of Duke's ECE 280 lab redesign, which modernizes classic analog and digital signal processing experiments for 21st-century embedded platforms. The emphasis on Arduino as a universal platform for across-the-curriculum integration reflects Duke's pedagogical goal of making signal processing accessible and immediately relevant to all ECE students.

References

- A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed. Pearson, 2010.
- J. H. McClellan, R. W. Schafer, and M. A. Yoder, *DSP First: A Multimedia Approach*, 2nd ed. Pearson, 2015.
- Arduino Official Documentation: `https://www.arduino.cc/reference/`
- Atmel ATmega328P Datasheet (Timer/Counter modes, PWM, interrupts).